

Fine-Grained Contrastive Learning for Robotic Surgery Skill Assessment Through Joint Instrument-Tissue Modeling

Abstract—In robot-assisted surgery, precise and objective surgical skill assessment is essential for ensuring patient safety, optimizing surgeon training, and advancing semi-autonomous surgical systems. However, existing automated evaluation methods generally depend on coarse, video-level global features, failing to capture the fine-grained nuances of surgical maneuvers. To address this, we propose a fine-grained contrastive learning framework that decomposes skill assessment into three critical dimensions: instrument manipulation, tissue handling, and surgical field clarity. Specifically, we extract video spatiotemporal features fused with instrument mask-guided attention maps to explicitly model dynamic instrument operations. Simultaneously, an intra-video contextual module encodes static global features to characterize tissue state changes and surgical field clarity. Cross-attention mechanisms are then introduced to capture discriminative inter-video representations across key assessment dimensions, and a contrastive regressor is employed to predict relative score disparities. We extensively validate the proposed model on a simulated dataset and two clinical datasets (laparoscopic cholecystectomy and hernia repair suturing). Our method achieves state-of-the-art performance in Mean Absolute Error (MAE) and Spearman’s rank correlation on both simulated and cholecystectomy datasets. Notably, on the hernia repair suturing dataset, it yields high-precision regression with a MAE of 0.98 (corresponding to 3.92% of the full GOALS score). Furthermore, we construct the largest surgical instrument mask dataset to date, comprising 201,318 frames from simulated and clinical videos, providing a valuable resource for future research.

Index Terms—Surgical skill assessment, Robot-assisted surgery, Fine-grained contrastive learning, Cross-attention mechanism.

I. INTRODUCTION

With hundreds of millions of surgeries performed worldwide each year, surgical proficiency is a key factor in operative success and is closely associated with patient outcomes [47]. Enhancing surgical skills can lead to better patient outcomes, including higher long-term survival rates and fewer postoperative complications [3]. Therefore, before entering clinical practice, surgical trainees must undergo systematic skills assessments conducted by board-certified surgeons to ensure their competency in real-world surgical procedures. Traditional methods for Surgical Skills Assessment (SSA) mainly rely on direct intraoperative observation or retrospective video review [32]. However, due to variability in evaluators’ standards, these approaches often lack consistency. To improve standardization, structured rating frameworks such as the Objective Structured Assessment of Technical Skills (OSATS) [25] have been introduced to evaluate surgical performance across multiple dimensions, as illustrated in Fig. 1(c). Although OSATS provides a more structured assessment process, it still depends on manual scoring and may be influenced by inter-rater variability and human bias. Given the increasing volume of surgical procedures and the growing demand for trainee

certification, there is an urgent need to develop accurate, automated, and reproducible surgical skills assessment systems that can improve evaluation efficiency, reduce inter-rater variability, and support consistent assessment outcomes. In recent years, robot-assisted surgery has been increasingly adopted in clinical practice due to its enhanced dexterity, improved visualization, and minimally invasive access. These systems routinely generate high-quality endoscopic videos and, in some platforms, synchronized kinematic data, providing valuable information for analyzing surgical performance. Building on these data sources, recent studies have explored deep learning-based methods for SSA, laying the foundation for automated and reproducible evaluation systems.

Early investigations on automated SSA in Robot-Assisted Minimally Invasive Surgery (RAMIS) have extensively explored kinematic data recorded from robotic surgical systems. For instance, Fawaz et al. [15] introduced a framework based on a fully convolutional neural network to classify proficiency levels—categorized as novice, intermediate, and expert—by processing high-dimensional motion trajectories. To further refine feature representation, Zia et al. [55] extracted multifaceted features, including texture, frequency, and entropy, employing a weighted feature fusion strategy to enhance classification robustness. More recently, Quarez et al. [29] proposed a weakly supervised paradigm utilizing a recursive cross-attention mechanism. This architecture effectively captures long-range temporal dependencies within complex surgical tasks, yielding significant performance gains with Spearman’s Rank Correlation Coefficients (SRCC) ranging from 0.83 to 0.88. Despite these advancements, kinematic-based paradigms encounter fundamental bottlenecks. Surgical maneuvers are inherently nonlinear, non-stationary, and characterized by high stochasticity, rendering low-level motion data insufficient for capturing the full complexity of surgical expertise [46]. While Zhang et al. [53] attempted to mitigate these issues by transforming one-dimensional kinematic series into two-dimensional time–frequency representations via continuous wavelet transform for vision transformer analysis, several critical limitations remain. First, kinematic metrics such as velocity and instrument pose primarily reflect human-in-the-loop manual dexterity rather than the underlying intelligence or autonomy of the robotic system [14], [53]. Notably, clinical indicators like “hand stability,” which typically correlate with surgeon experience, are often artificially enhanced in RAMIS by control-layer damping algorithms and motion filters, potentially masking true skill variability. Furthermore, the acquisition of high-fidelity kinematic data necessitates specialized robotic interfaces or extrinsic sensor integration, imposing substantial overhead in terms of system cost and deployment complexity.

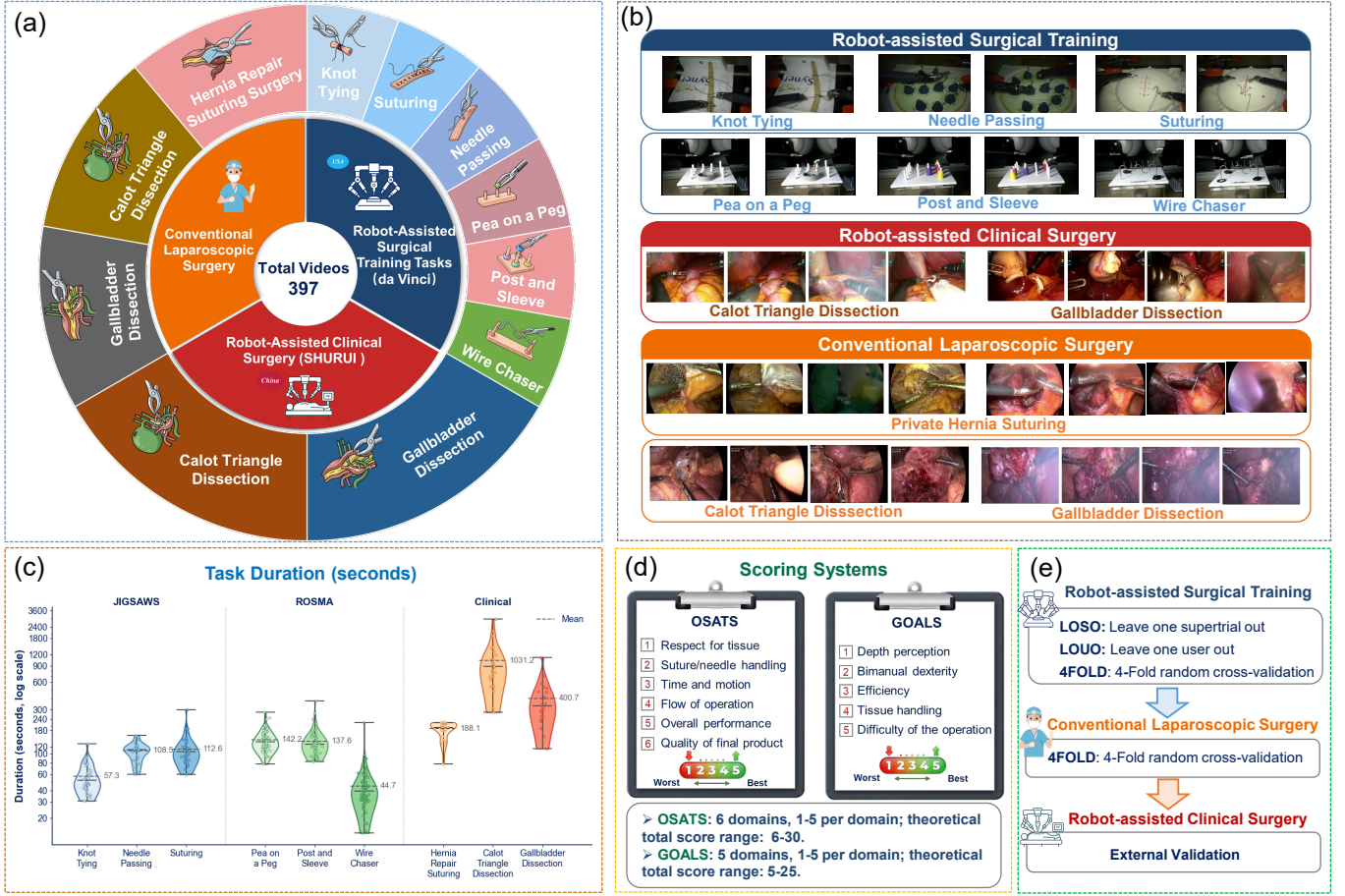


Fig. 1: Overview of the dataset composition, representative video frames, task duration distributions, scoring systems, and experimental settings. (a) summarizes the multi-source surgical skill assessment dataset used in this study. The dataset covers three surgical settings: robot-assisted surgical training, robot-assisted clinical surgery, and conventional laparoscopic clinical surgery, and includes nine tasks or surgical phases. (b) shows representative video frames from the different data sources, highlighting variations in visual appearance, task setup, and operative environment. (c) presents the task duration distributions, with dashed lines indicating the mean duration for each task. (d) summarizes the two surgical skill scoring systems used for annotation. (e) illustrates the experimental settings. Robot-assisted training tasks were evaluated using LOSO, LOUO, and randomized 4-fold cross-validation; conventional laparoscopic clinical surgery was evaluated using randomized 4-fold cross-validation; and robot-assisted clinical surgery was used as an independent dataset for external validation.

Unlike kinematic signals, intraoperative videos are inherently available in both conventional laparoscopic surgery and RAMIS. Consequently, video-based SSA has increasingly emerged as a pivotal research frontier in the field of surgical performance evaluation. Leveraging rapid advancements in deep learning, recent methodologies have successfully extracted high-dimensional and abstract spatiotemporal features, significantly augmenting the accuracy and robustness of SSA frameworks [18] [12] [28] [45]. Empirical evidence consistently suggests that video-based approaches offer superior performance over kinematic-based counterparts [31]. Currently, video-based SSA is primarily bifurcated into skill-level classification and continuous score regression. For skill-level classification, Funke et al. divided surgical videos into subsequences and integrated Inflated 3D Convolutional Networks (I3D) [6] and temporal segmentation networks to model three tasks in the JHU-ISI gesture and skill assessment working set

(JIGSAWS) [10], reporting classification accuracies between 95.1% and 100% [12]. Similarly, Rezaei et al. [31] coupled optical-flow convolutional neural networks (CNNs) with tree-based Gaussian process classifiers to enhance discriminative power. However, these classification paradigms typically rely on coarse, discrete labels (novice, intermediate, expert), which lack the granularity required to capture subtle technical nuances or procedure-specific variances. To achieve higher assessment fidelity, recent efforts have shifted toward continuous skill-score regression. Wang et al. [45] utilized LSTMs to capture temporal dependencies from visual features for score prediction, while Ding et al. proposed a local-global difference modeling framework to associate long-term surgical processes with localized events. Furthermore, Tang et al. introduced an uncertainty-aware score distribution learning approach based on 3D CNNs, in which skill scores were modeled as probability distributions to enhance prediction accuracy [33]. Liu

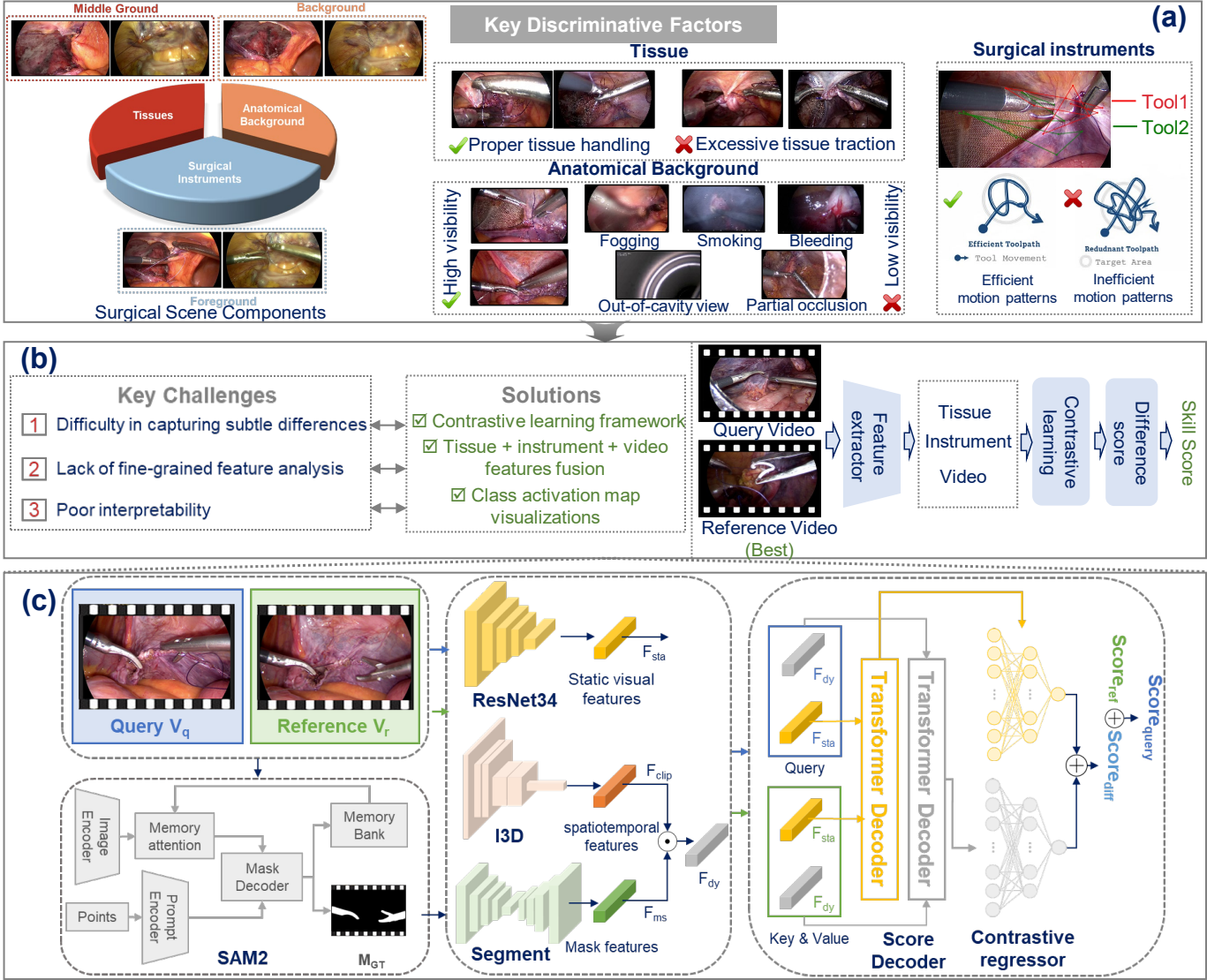


Fig. 2: Overview of the overall method framework. (a) The hierarchical visual composition of surgical videos and the three corresponding key skill assessment dimensions; (b) The primary challenges in current surgical skill assessment, the proposed strategies to address them, and the overall workflow; (c) A detailed schematic illustrating the architecture of the proposed fine-grained contrastive learning framework.

et al. developed a multi-path feature fusion model, achieving a mean Spearman correlation coefficient of 0.805 on the JIGSAWS dataset [22]. Majeedi et al. incorporated a deep probabilistic model for skill score prediction, which further improved regression accuracy by incorporating uncertainty modeling [24].

Although score regression provides a more refined characterization of performance than coarse classification, most existing methods treat surgical videos as independent samples, thereby failing to explicitly model the relative skill discrepancies between procedures. To address this gap, Anastasiou et al. introduced a reference-video-based contrastive regression strategy that computes the relative skill difference between a query video and a high-performing demonstration [1]. This comparison-centric formulation is particularly effective for SSA, as technical expertise is often more discernible through

contrastive analysis against "gold standard" performances. Nevertheless, surgical skill is not solely encapsulated by global, video-level spatiotemporal features. As shown in Fig. 2(a), surgical scenes are composed of foreground instruments, middle-ground tissue structures, and background anatomical contexts. These visual constituents map directly to critical clinical dimensions: instrument manipulation, tissue handling, and surgical field clarity [22]. Expert surgeons do not merely execute efficient trajectories; they actively manage the operative environment by mitigating obstructions (e.g., bleeding, smoke) and regulating interaction forces to prevent collateral tissue damage. Consequently, relying on holistic video features may overlook these fine-grained visual cues. [20] [5] [4].

Building upon the paradigm of reference-video-based contrastive regression [1], we propose a fine-grained visual modeling framework tailored for SSA in RAMIS. The pro-

posed framework extends conventional pairwise video comparisons by synergistically integrating instrument-aware guidance, tissue-centric contextual representations, and long-term temporal process modeling. Such a design empowers the model to discern expertise by leveraging complementary visual cues associated with instrument manipulation dexterity, dynamic tissue response, and procedural fidelity. The main contributions of this work are summarized as follows:

1. Construction of a Multi-Source Surgical Video Collection and Instrument Mask Annotation Set: We construct a large-scale multi-source surgical video collection for robot-assisted SSA. The collection includes robotic simulation training tasks, conventional laparoscopic clinical videos, and real robot-assisted clinical surgery videos. In addition, we provide 201,318 surgical instrument mask annotations generated using the Segment Anything Model v2 (SAM2) [30] and refined through manual verification.

2. Instrument-Region Motion Modeling with Segmentation-Guided Feature Reweighting: We introduce an I3D-based dynamic branch to extract snippet-level motion representations from surgical videos. To incorporate instrument-region information, an auxiliary segmentation branch is trained using the quality-controlled instrument masks. The resulting segmentation-derived features are used to reweight dynamic representations in the aligned feature space, guiding the model toward instrument-region motion cues without directly multiplying image-space masks with I3D feature maps.

3. Frame-Level Tissue-Context Representation for Surgical Scene Modeling: We introduce a ResNet-34-based tissue-context branch to extract frame-level appearance and surgical-scene contextual features. This branch complements instrument-region motion modeling by providing visual evidence related to tissue structures, instrument–tissue interaction regions, and surgical-field appearance.

4. Fine-Grained Contrastive Modeling of Instrument and Tissue-Context Features: Building on reference-video-based contrastive regression, we use cross-attention to model differences between query and reference videos across dynamic instrument features and frame-level contextual features. The joint modeling of feature differences and score differences supports continuous skill-score prediction and video-level ranking consistency.

We evaluate the proposed framework on an expanded and heterogeneous dataset collection, as illustrated in Fig. 1. The evaluation includes the publicly available JIGSAWS, ROSMA [41], and HeiChole [44] datasets, as well as two private datasets: a hernia repair suturing dataset and a SHURUI robot-assisted cholecystectomy dataset. This evaluation covers three surgical settings: robot-assisted surgical training tasks using the da Vinci platform, conventional laparoscopic surgery, and robot-assisted clinical surgery using the SHURUI surgical platform, enabling broader assessment across different surgical platforms, operative tasks, and clinical scenarios.

II. METHODS

As delineated in Fig. 2(b), current automated Surgical Skill Assessment (SSA) methodologies encounter two main

limitations: first, conventional single-video paradigms often lack the sensitivity required to discern subtle inter-procedural technical variances; second, global video-level representations may overlook local skill-related visual cues. To mitigate these limitations, we build on reference-video-based contrastive regression and introduce a fine-grained instrument–tissue visual modeling framework. The framework integrates an I3D-based motion branch, a segmentation-guided feature reweighting module, and a ResNet-34-based tissue-context branch. Furthermore, we employ CAM [54] as a post hoc qualitative visualization tool. This allows us to elucidate the spatial regions prioritized by disparate feature branches during inference. Fig. 2(c) illustrates the comprehensive architecture of the proposed Fine-grained instrument–tissue contrastive learning framework. An I3D backbone extracts snippet-level spatiotemporal dynamics, while a ResNet-34-based branch captures frame-level appearance descriptors pertinent to tissue morphology and surgical-scene context. In parallel, the segmentation-guided feature reweighting module uses quality-controlled instrument-mask annotations to train an auxiliary segmentation branch, whose segmentation-derived features reweight the dynamic representations in the aligned feature space and guide the model toward instrument-region motion cues. After obtaining these complementary representations, the framework models query-reference differences across instrument-region motion features and tissue-context features. Finally, contrastive score regressors map the branch-specific difference representations to relative score differences, which are combined to support continuous skill-score prediction and video-level ranking consistency.

A. Problem Formulation

Consider a surgical video $V \in \mathbb{R}^{T \times H \times W \times C}$ annotated with a skill score y , where T , H , W , and C denote the clip length, height, width, and number of channels, respectively. Video-based SSA is commonly formulated as a direct regression problem, where a video is used as input to predict its corresponding skill score:

$$\hat{y} = \mathcal{R}(\mathcal{E}(V)) \quad (1)$$

where $\mathcal{E}(\cdot)$ and $\mathcal{R}(\cdot)$ denote the feature extractor and the regression network, respectively. However, holistic video-based regression methods exhibit limitations in capturing subtle distinctions between surgical videos, especially when the videos have similar annotated scores. To address this limitation, we build on reference-video-based contrastive regression [1] and propose a fine-grained extension that formulates SSA as the prediction of the relative skill-score difference between a query video $V_q \in \mathbb{R}^{T \times H \times W \times C}$ and a reference video $V_r \in \mathbb{R}^{T \times H \times W \times C}$. Specifically, each input consists of a video pair, with one serving as the query video V_q and the other as a reference video V_r associated with an annotated skill score y_r . By comparing complementary visual representations related to instrument-region motion, tissue-related scene context, and procedural dynamics between the query and reference videos, the model learns discriminative relative differences associated

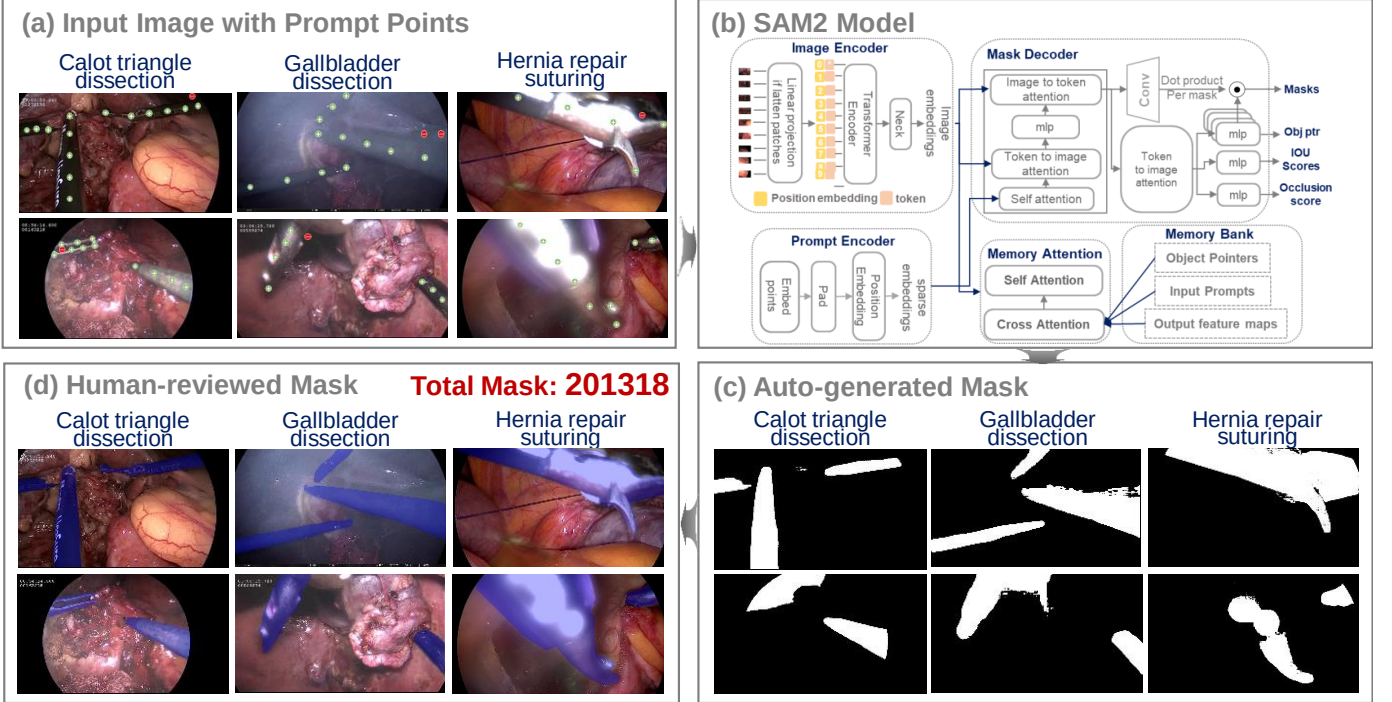


Fig. 3: Overview of the workflow for constructing the largest SSA-oriented surgical instrument mask dataset using SAM2. (a) Original video frames accompanied by corresponding point-based prompts; (b) Schematic illustration of the SAM2 model architecture; (c) Instrument masks automatically generated by SAM2; (d) Overlay of manually refined instrument segmentation masks on the original video frames.

with surgical skill. This regression task is formally expressed as:

$$y_q = \mathcal{CR}(\mathcal{E}(V_q), \mathcal{E}(V_r)) + y_r \quad (2)$$

where $\mathcal{CR}$ denotes the contrastive regressor. During the training phase, V_q and V_r are used as inputs, the skill score difference between the two surgical videos is regressed, and the network parameters are optimized by minimizing the discrepancy between the predicted and ground-truth score differences.

B. High-Quality Surgical Instrument Mask Dataset Construction via SAM2

Driven by the proliferation of promptable foundation models, Segment Anything Model (SAM) [39] and its successor, SAM2 [30], have introduced versatile paradigms for interactive mask generation. Building upon these, several surgical-domain adaptations have investigated their efficacy in surgical video segmentation [35]–[37]. As a large-scale model designed for general-purpose segmentation, SAM2 exhibits robust zero-shot performance across diverse visual domains. In this work, we employ SAM2 as an offline initialization engine for instrument-mask annotation, complemented by a rigorous phase of manual verification and refinement. By leveraging this human-in-the-loop pipeline, we constructed a high-fidelity surgical instrument segmentation dataset that spans both simulated environments and clinical scenarios. Comprising 201,318 annotated frames, this represents the most extensive instrument mask repository specifically curated for

SSA to date. As delineated in Fig. 3(b), the SAM2 architecture integrates five pivotal modules: the Image Encoder, Prompt Encoder, Mask Decoder, Mask Attention mechanism, and Memory Bank.

As illustrated in Fig. 3, SAM2 is employed as an offline initialization engine to generate preliminary surgical instrument masks. For each video sequence, an interactive prompting procedure is executed on the initial frame, where positive points are sampled within instrument regions, and negative points are placed in the background. These prompts, along with the video stream, are processed by SAM2 to yield initial mask candidates. To ensure high-fidelity annotations, we perform a manual refinement phase to rectify instances of segmentation failure, such as missing regions, false positives, or imprecise boundaries [see Fig. 3 (c)]. This rigorous quality control pipeline results in the finalized instrument masks shown in Fig. 3 (d). To quantify the annotation reliability and the extent of human intervention, we conducted a quality audit by randomly sampling video segments from various surgical tasks. We report two primary metrics: the manual correction rate and the Dice similarity coefficient between the raw SAM2 output and the post-refinement masks. For simulation tasks, the manual correction rate ranged from 2.34% to 5.36%, with a corresponding Dice score of 89.36%–93.49%. In contrast, for more challenging clinical scenarios, the correction rate increased to 7.14%–9.87%, with a Dice score of 86.52%–88.37%. These quality-controlled masks serve as instrument-region spatial priors to guide the auxiliary segmentation branch. The resulting segmentation-derived features are used

to reweight dynamic representations for subsequent query-reference comparison.

C. Instrument-Region Motion Feature Modeling

1) *Short-Term Spatiotemporal Feature Extraction:* To model short-term surgical motion patterns, the query video V_q and the reference video V_r are processed by the same I3D-based feature extraction pipeline. Before snippet construction, each video is uniformly sampled according to the dataset-specific preprocessing protocol; therefore, the raw frame rates of different datasets are not directly used to define the model input. Each sampled video sequence is then partitioned into N overlapping snippets $\{X_i\}_{i=1}^N$ using a sliding window with a stride of 10 sampled frames. Each snippet $X_i \in \mathbb{R}^{C \times 16 \times H \times W}$ contains 16 consecutive sampled frames, where C , H , and W denote the number of channels, height, and width, respectively. This overlapping strategy helps preserve local temporal continuity between adjacent snippets. Let $\mathcal{E}_{dy}(\cdot)$ denote the I3D-based dynamic feature extractor. For each snippet X_i , the compact dynamic representation is computed as:

$$F_{clip}(X_i) = \mathcal{E}_{dy}(X_i). \quad (3)$$

where $F_{clip}(X_i) \in \mathbb{R}^{D_d}$ denotes the compact spatiotemporal feature extracted from the i -th snippet, and D_d denotes the feature dimension. The features from all N snippets are then arranged in temporal order to form a window-level sequence representation, which is used for subsequent segmentation-guided feature reweighting and query-reference comparison.

2) Segmentation-Guided Feature Reweighting Module:

To incorporate instrument-region information, we introduce an auxiliary segmentation branch trained with the quality-controlled instrument masks generated during dataset preparation. SAM2 is used only for offline mask initialization and annotation construction, and is not used as the segmentation model during skill assessment. During both training and inference, the query and reference videos are concatenated and divided into the same 16-frame overlapping snippets as the dynamic feature branch. These snippets are fed into the segmentation branch, which outputs multi-scale instrument-mask predictions and a compact segmentation-derived feature used for dynamic feature reweighting. Let $F_{ms}(X_i)$ denote the compact segmentation-derived feature extracted from snippet X_i . The segmentation branch is supervised by the quality-controlled instrument-mask annotations. For each snippet, the segmentation branch produces K multi-scale mask predictions $\{\hat{M}_i^{(k)}\}_{k=1}^K$, where $K = 5$ in our implementation. These predictions are optimized using focal loss:

$$\mathcal{L}_{mask} = \sum_{k=1}^K \mathcal{L}_{focal}(\hat{M}_i^{(k)}, M_i), \quad (4)$$

where M_i denotes the quality-controlled instrument mask corresponding to snippet X_i . This loss trains the auxiliary segmentation branch to encode instrument-region information. The masks are used as spatial priors for instrument-region modeling, rather than as skill-score labels. After the segmentation-derived features are obtained for all snippets,

they are arranged in the same temporal order as the I3D dynamic features. Let $F_{clip}(V)$ denote the window-level dynamic feature sequence extracted by the I3D branch, and let $F_{ms}(V)$ denote the corresponding segmentation-derived feature sequence. Before fusion, $F_{ms}(V)$ is reshaped to match the window-level structure of $F_{clip}(V)$. A sigmoid activation function is then applied to obtain continuous-valued weighting coefficients:

$$\tilde{F}_{ms}(V) = \sigma(F_{ms}(V)). \quad (5)$$

The segmentation-guided dynamic representation is computed as:

$$F_{dy}(V) = F_{clip}(V) \odot \tilde{F}_{ms}(V), \quad (6)$$

where $\odot$ denotes element-wise multiplication. This operation is performed in the aligned feature space, rather than by directly multiplying image-space masks with I3D feature maps. The resulting representation provides instrument-region guidance for subsequent query-reference comparison.

D. Intra-Video Context Modeling Module

In addition to instrument-region motion cues, tissue-related scene context and surgical-field appearance provide complementary visual information for video-based SSA. Therefore, we introduce an intra-video context modeling module to extract frame-level appearance and surgical-scene contextual features as a complement to the I3D-based dynamic branch. Accordingly, ResNet-34 [13] is adopted as the backbone of this branch. While the I3D branch captures spatiotemporal dynamics from video snippets, the ResNet-34 branch extracts frame-level appearance and contextual cues from individual frames, including information related to tissue-interaction regions and surgical-field appearance. These contextual cues provide complementary information for fine-grained SSA.

Given a sampled video frame x_t , the ResNet-34 module $\mathcal{R}(\cdot)$ extracts a frame-level contextual feature:

$$F_{ctx}(x_t) = \mathcal{R}(x_t). \quad (7)$$

The frame-level features from the sampled video are then arranged in temporal order and processed by a contextual projection module $\mathcal{D}_{ctx}(\cdot)$ to obtain the intra-video contextual representation:

$$F_{sta} = \mathcal{D}_{ctx}(\{F_{ctx}(x_t)\}_{t=1}^T). \quad (8)$$

Here, F_{sta} denotes the tissue-context representation used for subsequent query-reference comparison. This branch does not explicitly estimate tissue state or surgical-field clarity as separate supervised outputs; instead, it provides frame-level contextual appearance features that complement the instrument-region motion representation.

E. Fine-Grained Contrastive Learning Framework

Surgical videos may share similar workflows or close annotated scores while still differing in local skill-related patterns, such as instrument-region motion, tissue interaction, and procedural execution. Most existing approaches predict skill scores from individual videos independently [8], [51],

which may limit their ability to model relative differences between procedures. Building on reference-video-based contrastive regression [1], we use a fine-grained query-reference contrastive modeling framework to compare a query video with a reference video and estimate their relative skill-score difference.

The framework consists of contrastive decoders and contrastive score regressors. Specifically, the instrument-region motion features extracted by the segmentation-guided dynamic branch and the tissue-context features extracted by the intra-video context branch are compared separately between the query and reference videos. The resulting branch-specific difference representations are then fed into score regressors to predict the relative score difference. The predicted score of the query video is obtained by adding the predicted score difference to the annotated score of the reference video.

1) *Contrastive Score Decoder*: Let F_{dy}^q and F_{dy}^r denote the instrument-region motion features of the query and reference videos, respectively. Let F_{sta}^q and F_{sta}^r denote their corresponding tissue-context features. To preserve the complementary information provided by these two feature types, we use separate contrastive decoders for the dynamic branch and the contextual branch.

For the dynamic branch, the query feature F_{dy}^q is used as the query input, while the reference feature F_{dy}^r is used as the key and value input. The cross-attention operation is formulated as:

$$Q_{dy} = F_{dy}^q \mathcal{W}_Q^{dy}, \quad K_{dy} = F_{dy}^r \mathcal{W}_K^{dy}, \quad V_{dy} = F_{dy}^r \mathcal{W}_V^{dy}, \quad (9)$$

$$\text{Atten}(Q_{dy}, K_{dy}, V_{dy}) = \text{softmax} \left(\frac{Q_{dy} K_{dy}^T}{\sqrt{d_k}} \right) V_{dy}, \quad (10)$$

where $\mathcal{W}_Q^{dy}$, $\mathcal{W}_K^{dy}$, and $\mathcal{W}_V^{dy}$ are learnable projection matrices, and d_k denotes the dimension of the key vectors. The scaling factor $\sqrt{d_k}$ is used to stabilize the magnitude of the dot-product attention scores.

The multi-head cross-attention output for the dynamic branch is computed as:

$$\text{MHA}_{dy} = \text{Concat}(\text{head}_1, \dots, \text{head}_n) \mathcal{W}_O^{dy}, \quad (11)$$

where n denotes the number of attention heads and $\mathcal{W}_O^{dy}$ is the output projection matrix. In this study, $n = 8$ is used unless otherwise specified. The resulting output is denoted as:

$$F_{dy}^{diff} = \mathcal{D}_{dy}(F_{dy}^q, F_{dy}^r). \quad (12)$$

The same operation is applied to the tissue-context branch:

$$F_{sta}^{diff} = \mathcal{D}_{sta}(F_{sta}^q, F_{sta}^r). \quad (13)$$

Here, F_{dy}^{diff} and F_{sta}^{diff} denote the query-reference difference representations for instrument-region motion features and tissue-context features, respectively.

2) *Contrastive Score Regressor*: The difference representations from the two branches are mapped to branch-specific score differences using two score regressors:

$$\hat{\Delta}y_{dy} = \mathcal{CR}_{dy}(F_{dy}^{diff}), \quad (14)$$

$$\hat{\Delta}y_{sta} = \mathcal{CR}_{sta}(F_{sta}^{diff}). \quad (15)$$

The final predicted score difference is obtained by averaging the branch-specific predictions:

$$\hat{\Delta}y_{q,r} = \frac{1}{2} (\hat{\Delta}y_{dy} + \hat{\Delta}y_{sta}). \quad (16)$$

The predicted score of the query video is then computed as:

$$\hat{y}_q = y_r + \hat{\Delta}y_{q,r}, \quad (17)$$

where y_r denotes the annotated score of the reference video.

During training, the model is optimized by minimizing the discrepancy between the predicted score difference and the annotated score difference:

$$\mathcal{L}_{score} = \left\| \hat{\Delta}y_{q,r} - (y_q - y_r) \right\|_2^2. \quad (18)$$

In implementation, query-to-reference and reference-to-query directions are both used during training to encourage consistency in relative score prediction.

Overall, the proposed contrastive framework models relative differences between query and reference videos using instrument-region motion features and tissue-context features. This design supports continuous skill-score prediction and video-level ranking consistency, without claiming that the model explicitly estimates tissue state changes or surgical-field clarity as separate supervised outputs.

III. EXPERIMENTAL SETUP

A. Datasets

1) *Robot-Assisted Simulation Training Datasets Using the da Vinci Platform*: To evaluate the proposed framework in robot-assisted surgical training scenarios, we included two datasets collected using da Vinci-based robotic platforms: JIGSAWS and ROSMA. JIGSAWS [10] is a widely used public benchmark for SSA. The dataset was collected using the da Vinci surgical system and contains three representative robotic training tasks: Knot Tying (KT), Suturing (SU), and Needle Passing (NP). Eight participants with different experience levels were included, consisting of two experts, two intermediates, and four novices. Each participant performed five trials for each task, resulting in 36 KT videos, 39 SU videos, and 28 NP videos. As shown in Fig. 1, the average durations of KT, SU, and NP were 57.3 s, 112.6 s, and 108.6 s, respectively. The videos were annotated using a modified OSATS protocol, and the observed total score ranges were [6, 22] for KT, [8, 30] for SU, and [7, 24] for NP.

ROSMA [41] is another robot-assisted simulation training dataset collected using the da Vinci Research Kit. It includes three training tasks: Post and Sleeve, Pea on a Peg, and Wire Chaser, performed by 12 participants. The dataset contains 206 videos, including 65 Post and Sleeve videos, 71 Pea on a Peg videos, and 70 Wire Chaser videos. The average durations of these tasks were 137.6 s, 142.2 s, and 44.7 s, respectively. Similar to JIGSAWS, ROSMA was evaluated using a modified OSATS scoring protocol. To obtain expert annotations consistent with this study, two clinically experienced surgeons independently re-annotated the ROSMA videos, and score discrepancies were resolved through consensus discussion. The observed total score ranges for Post and Sleeve, Pea on

TABLE I: Quantitative comparison results on the Simulated dataset. The red marker represents the best results.

Input	Method	KT			NP			SU			Avg.		
		LOSO	LOUO	4-FOLD	LOSO	LOUO	4-FOLD	LOSO	LOUO	4-FOLD	LOSO	LOUO	4-FOLD
SRCC↑													
K	SMT-DCT-DFT [55]	0.70	0.73	-	0.38	0.23	-	0.64	0.10	-	0.59	0.40	-
	DCT-DFT-ApEn [55]	0.63	0.60	-	0.46	0.25	-	0.75	0.37	-	0.63	0.42	-
	Recap [29]	0.88	-	-	0.85	-	-	0.83	-	-	0.85	-	-
K+V	AIM [9]	-	-	0.82	-	-	0.65	-	-	0.63	-	-	0.70
	MultiPath-VTP [22]	-	0.58	0.78	-	0.62	0.76	-	0.45	0.79	-	0.55	0.78
	MultiPath-VTPE [22]	-	0.59	0.82	-	0.65	0.76	-	0.45	0.83	-	0.57	0.81
V	ResNet-LSTM [45]	0.52	0.36	-	0.84	0.33	-	0.73	0.67	-	0.72	0.47	-
	NETS [2]	-	-	0.85	-	-	0.80	-	-	0.86	-	-	0.84
	Relation Graphs [26]	-	-	0.75	-	-	0.54	-	-	0.36	-	-	0.57
	USDL [33]	-	-	0.61	-	-	0.63	-	-	0.64	-	-	0.63
	MUSDL [33]	-	-	0.71	-	-	0.69	-	-	0.71	-	-	0.70
	Core [49]	-	-	0.86	-	-	0.86	-	-	0.84	-	-	0.85
	Pan et al. [27]	-	-	0.84	-	-	0.75	-	-	0.75	-	-	0.78
	multitask [16]	0.77	-	-	0.72	-	-	0.76	-	-	0.75	-	-
	*S3D [48]	0.64	0.14	-	0.57	0.35	-	0.68	0.03	-	0.63	0.18	-
	*ResNet-MTL-VF [45]	0.63	0.72	-	0.73	0.48	-	0.79	0.68	-	0.72	0.64	-
	*C3D-MTL-VF [45]	0.89	0.83	-	0.75	0.86	-	0.77	0.69	-	0.81	0.80	-
	ViSA [21]	0.92	0.76	0.84	0.93	0.90	0.86	0.84	0.72	0.79	0.90	0.81	0.83
	S ⁴ AQA [52]	-	-	0.81	-	-	0.55	-	-	0.53	-	-	0.66
	Contra-Sformer [1]	0.89	0.69	0.87	0.71	0.71	0.81	0.86	0.65	0.69	0.83	0.68	0.81
	DAE [51]	-	-	0.87	-	-	0.86	-	-	0.86	-	-	0.86
	ZEAL [19]	-	-	0.54	-	-	0.67	-	-	0.62	-	-	0.61
	TRSA [50]	-	-	0.75	-	-	0.81	-	-	0.70	-	-	0.75
	MT-VIT [11]	0.93	0.85	0.89	0.86	0.88	0.82	0.87	0.75	0.82	0.89	0.83	0.84
	Ours	0.91	0.85	0.89	0.92	0.94	0.89	0.89	0.73	0.86	0.91	0.84	0.88
MAE ↓													
K	Recap [29]	2.04	-	-	3.12	-	-	2.89	-	-	2.68	-	-
V	ViSA [21]	2.16	2.01	2.60	1.66	2.16	2.05	2.58	2.82	2.70	2.13	2.33	2.45
	Contra-Sformer [1]	1.75	1.39	2.10	3.15	3.17	3.21	2.74	2.58	2.99	2.55	2.38	2.77
	MT-VIT [11]	1.69	1.45	2.00	1.95	2.04	2.23	2.23	2.46	2.43	1.95	1.98	2.22
	Ours	1.68	1.36	1.92	1.62	1.89	1.81	2.45	2.08	2.79	1.92	1.78	2.17

a Peg, and Wire Chaser were [11, 27], [6, 28], and [7, 29], respectively. Together, JIGSAWS and ROSMA constitute the robot-assisted simulation training category in this study. Both datasets were collected using da Vinci-based robotic platforms and cover different fundamental training tasks, task durations, and score distributions. They therefore provide a controlled robotic training setting for evaluating the generalization ability of the proposed framework in robot-assisted surgical training scenarios.

Following the evaluation protocols commonly adopted for JIGSAWS, we evaluated both JIGSAWS and ROSMA under three settings: Leave-One-User-Out (LOUO), Leave-One-Supertrial-Out (LOSO), and randomized 4-fold cross-validation. These complementary protocols assess different aspects of generalization in robot-assisted surgical training scenarios, including user-level generalization, trial-level robustness, and overall predictive performance. In the LOUO setting, all samples from one participant are held out for

testing, while samples from the remaining participants are used for training. This protocol evaluates whether the model can generalize to unseen users. For JIGSAWS, LOUO results in an 8-fold evaluation, corresponding to its eight participants. The same user-level evaluation is applied to ROSMA across its 12 participants. In the LOSO setting, all samples corresponding to one repetition index are held out for testing, while samples from the remaining repetitions are used for training. This protocol evaluates robustness across repeated trials of the same task. For JIGSAWS, LOSO results in a 5-fold evaluation because each task was repeated five times. For ROSMA, the same supertrial-based evaluation is applied according to its repetition structure. In addition, a randomized 4-fold cross-validation is conducted to evaluate overall predictive performance under a commonly used data-splitting strategy. Together, these three protocols provide a consistent basis for evaluating the proposed framework across the two robot-assisted surgical training datasets.

2) *Robot-Assisted Clinical Surgery Dataset Using the SHURUI Platform:* We further constructed a private robot-assisted cholecystectomy dataset collected using the SHURUI surgical robotic platform in China. This dataset was introduced to evaluate the applicability of the proposed framework in real robot-assisted clinical surgery. Following the phase definitions adopted in HeiChole, each procedure was divided into two key surgical phases: Calot Triangle Dissection and Gallbladder Dissection. The dataset contains three complete robot-assisted cholecystectomy videos, with total procedure durations of 50 min 11 s, 46 min 20 s, and 48 min 24 s, respectively. Phase annotation was performed by experienced surgeons. The durations of the Calot Triangle Dissection phase were 28 min 38 s, 20 min 17 s, and 20 min 45 s for the three cases, respectively, while the corresponding durations of the Gallbladder Dissection phase were 21 min 33 s, 26 min 23 s, and 27 min 39 s. For skill scoring, we adopted a modified Global Operative Assessment of Laparoscopic Skills (GOALS) protocol consistent with the scoring framework used in HeiChole [43]. Two clinically experienced surgeons independently scored the videos, and any score discrepancies were resolved through consensus discussion. The final skill scores in the private SHURUI dataset ranged from 14 to 21. Given the limited availability of real robot-assisted clinical surgery data, this dataset was not used for model training. Instead, it served as an external validation set. Specifically, the model weights trained on HeiChole were directly applied to the SHURUI dataset without additional fine-tuning, thereby evaluating the cross-platform generalization of the proposed framework from conventional laparoscopic cholecystectomy to real robot-assisted cholecystectomy.

3) *Conventional Laparoscopic Surgery Datasets:* Conventional laparoscopic surgery and RAMIS both rely on endoscopic visual observation. Therefore, in addition to robot-assisted training data and robot-assisted clinical surgery data, we incorporated two conventional laparoscopic clinical datasets as supplementary validation datasets: the public HeiChole dataset and our private hernia repair suturing dataset. These datasets were included to evaluate the proposed method across different laparoscopic surgical scenarios. The HeiChole dataset [44] contains 24 complete laparoscopic cholecystectomy videos collected from three surgical centers in Germany. Each video records the full procedure, from laparoscope insertion into the abdominal cavity to final withdrawal. Following prior video-based SSA studies [1], [22], [33], we focused on two key phases of cholecystectomy: Calot Triangle Dissection and Gallbladder Dissection. These two phases yielded 48 phase-level samples for skill assessment. As shown in Fig. 1, the average durations of Calot Triangle Dissection and Gallbladder Dissection were 1031.2 s and 400.7 s, respectively, and the observed skill scores ranged from 15 to 21.

The private hernia repair suturing dataset contains 34 laparoscopic suturing videos collected from clinical procedures. Since this dataset was also collected in a laparoscopic surgical setting, we adopted a modified GOALS scoring protocol [43] for skill annotation. Two senior surgeons independently scored all videos, and samples with score discrepancies were reviewed through consensus discussion to obtain final scores. As shown

in Fig. 1, the average video duration was 188.1 s, and the observed skill scores ranged from 15 to 21. Overall, HeiChole and the private hernia repair suturing dataset constitute the conventional laparoscopic surgery category in this study. These datasets were not collected from robotic surgical platforms; rather, they provide supplementary clinical laparoscopic scenarios for evaluating model performance across different surgical procedures, video durations, and clinical settings.

B. Evaluation Metrics

We evaluated the proposed framework using two complementary metrics: Mean Absolute Error (MAE) and Spearman's Rank Correlation Coefficient (SRCC). MAE quantifies the score-level prediction error between the predicted and true skill scores, whereas SRCC measures the rank-order consistency between them. Therefore, MAE reflects regression accuracy, while SRCC evaluates whether the predicted scores preserve the relative ordering of surgical performance across videos.

1) *Spearman Rank Correlation Coefficient:* Following prior SSA studies, SRCC was utilized to evaluate the ranking consistency of the predicted skill scores. Let p_i and q_i denote the ranks of the predicted score and the true score for the i -th sample, respectively. Let $\bar{p}$ and $\bar{q}$ denote the corresponding mean ranks. SRCC is computed as the Pearson correlation coefficient between the two rank vectors:

$$SRCC = \frac{\sum_{i=1}^n (p_i - \bar{p})(q_i - \bar{q})}{\sqrt{\sum_{i=1}^n (p_i - \bar{p})^2 \sum_{i=1}^n (q_i - \bar{q})^2}} \quad (19)$$

The value of SRCC ranges from -1 to 1 , where a higher value indicates a stronger positive correlation and consequently better model performance.

2) *Mean Absolute Error:* MAE was used to quantify the discrepancy between predicted and ground-truth scores. The average magnitude of prediction errors is computed by taking the mean of the absolute differences between predicted and true scores across the test set. Given the predicted skill score $\hat{y}_i$ and the annotated skill score y_i for the i -th test sample, MAE is defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (20)$$

MAE ranges from 0 to $+\infty$, with smaller values indicating lower prediction error. Compared with SRCC, which evaluates ranking consistency, MAE provides a direct measure of the regression accuracy of continuous skill-score prediction.

C. Implementation Details

The proposed framework was implemented in PyTorch. Model training was conducted on a workstation equipped with an NVIDIA RTX 5090 GPU. The dynamic branch employed an I3D backbone [6] pre-trained on Kinetics [17] to extract snippet-level spatiotemporal representations. Additionally, the static visual encoder uses ResNet-34 as its backbone. During training, the initial learning rate was set to 1×10^{-4} for the backbone network and 1×10^{-3} for the tissue-related feature extraction module. The model was optimized using Adam with the weight decay set to 0 . The number of attention heads in

the cross-attention decoders was fixed at eight. The maximum number of training epochs was set to 100 for the robot-assisted simulation datasets and 150 for the clinical datasets. An early stopping strategy with a patience of 10 epochs was adopted to reduce overfitting. All input video frames were uniformly resized to 112×112 , and data augmentation techniques, including random horizontal flipping and rotation, were applied during training to enhance the model's generalization ability. The proposed framework contains 49.2 M trainable parameters, with a peak GPU memory consumption of 4.73 GB during inference. The model achieved an inference speed of 21 FPS, and the saved checkpoint size was 452.57 MB.

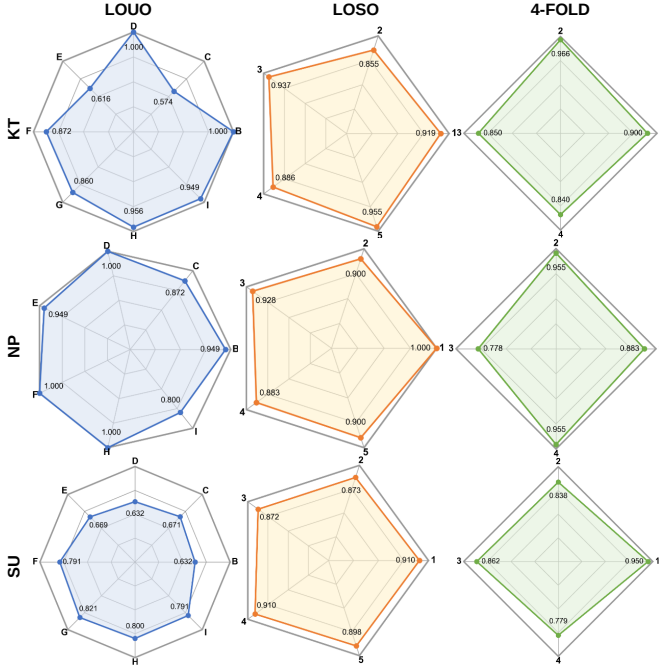


Fig. 4: Visualization of the SRCC result distributions for the three subtasks of the simulated dataset under different experimental paradigms using radar charts.

IV. RESULTS

A. Results on Robot-Assisted Simulation Training Datasets

1) *Results on the JIGSAWS Dataset:* Table I summarizes the MAE and SRCC results of the proposed framework on the JIGSAWS dataset, covering three robot-assisted training tasks, namely KT, NP, and SU, under LOUO, LOSO, and randomized 4-fold cross-validation. Overall, the model maintained favorable ranking consistency across most experimental settings. Except for SU under the LOUO setting, all SRCC values were higher than 0.80. In particular, the model showed stable ranking performance on NP and KT, with the highest SRCC reaching 0.94, indicating strong agreement between the predicted scores and expert-annotated scores. Compared with KT and NP, SU showed relatively lower SRCC values. This may be associated with the higher procedural complexity of suturing, which typically involves continuous needle manipulation, coordinated bimanual control, and more complex

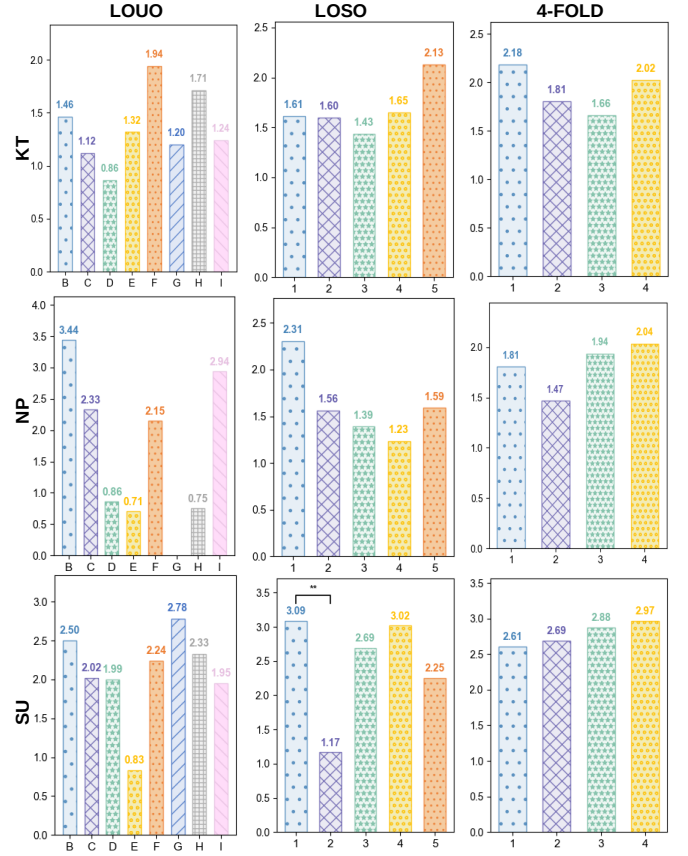


Fig. 5: Visualization of MAE metrics for the Simulated dataset across three experimental paradigms, where ** denotes statistically significant differences between the two-fold results.

instrument–tissue interactions. These factors may increase inter-user variability and make rank prediction more challenging under unseen-user or different data-splitting conditions. Nevertheless, the SU task still maintained acceptable ranking consistency in most settings. Fig. 4 further visualizes the fold-wise SRCC variations across different tasks and evaluation settings using radar charts. Fig. 5 presents the fold-wise MAE results across different tasks and evaluation settings. In the JIGSAWS experiments, the maximum MAE was 2.79, and the overall MAE range across tasks and evaluation settings was 1.43, corresponding to 4.77% of the maximum OSATS score of 30. These results indicate that the model maintained relatively low score-level prediction errors in continuous skill-score regression. Further analysis of fold-wise MAE variability showed that the MAE ranges across the three tasks were 0.56, 0.26, and 0.71, respectively. Across LOUO, LOSO, and randomized 4-fold cross-validation, the corresponding MAE ranges were 0.83, 0.72, and 0.98, respectively. These results suggest that the prediction errors remained relatively stable across folds, while the degree of variability was still affected by task type and evaluation protocol. Fold-level statistical comparisons showed a significant difference only between the first and second folds of SU under the LOSO setting, whereas no significant fold-level differences were observed in the other tasks or settings. Overall, the JIGSAWS results indicate

TABLE II: Performance of the proposed framework on the ROSMA dataset under randomized 4-fold cross-validation, LOUO, and LOSO evaluation settings.

	Post and Sleeve		Pea on a Peg		Wire Chaser		Avg.	
	MAE	SRCC	MAE	SRCC	MAE	SRCC	MAE	SRCC
4-Fold	2.837	0.625	2.834	0.684	2.358	0.799	2.676	0.703
LOUO	2.315	0.729	2.236	0.704	2.365	0.742	2.305	0.725
LOSO	2.480	0.725	2.988	0.701	2.505	0.743	2.658	0.723
Avg.	2.544	0.693	2.686	0.696	2.409	0.761	2.546	0.717

that the proposed query-reference contrastive framework can provide consistent continuous score prediction and ranking agreement on representative robot-assisted surgical training tasks.

2) *Results on the ROSMA Dataset:* Table II summarizes the MAE and SRCC results of the proposed method on the ROSMA dataset, covering three robot-assisted training tasks: Post and Sleeve, Pea on a Peg, and Wire Chaser. The model was evaluated under three protocols: randomized 4-fold cross-validation, LOUO, and LOSO. Overall, the proposed method showed stable continuous score prediction and rank-order consistency on ROSMA, with an average MAE of 2.546 and an average SRCC of 0.717. Across the three evaluation protocols, the best average performance was obtained under LOUO, with an MAE of 2.305 and an SRCC of 0.725. This result suggests that the model can maintain stable prediction performance when tested on unseen participants. Under LOSO and randomized 4-fold cross-validation, the average SRCC values were 0.723 and 0.703, respectively, indicating comparable ranking performance across different data-splitting strategies. At the task level, Wire Chaser achieved the highest average SRCC of 0.761 and reached 0.799 under randomized 4-fold cross-validation, suggesting favorable rank-order prediction for this task. Overall, the ROSMA results further support the applicability of the proposed framework to robot-assisted simulation training scenarios. Compared with JIGSAWS, ROSMA includes different training tasks, participants, and procedural patterns, thereby providing additional evidence for evaluating the proposed framework on robot-assisted surgical training data.

B. Results on Conventional Laparoscopic Surgery Datasets

1) *Results on the HeiChole Dataset:* To evaluate the proposed method in clinical laparoscopic cholecystectomy, we conducted experiments on the HeiChole dataset. Following the phase definition used in previous video-based SSA studies [7], [21], [23], each procedure was analyzed in two key phases: Calot Triangle Dissection and Gallbladder Dissection. Randomized 4-fold cross-validation was performed independently for each phase. As shown in Table III, the proposed method achieved an overall mean MAE of 0.97 and a mean SRCC of 0.76 across the two phases. As shown in Fig. 6(a), the MAE values on HeiChole ranged from 0.752 to 1.115, with an overall range of 0.363. Considering that the modified GOALS protocol used in this dataset has a theoretical total score range of 5–25, the obtained MAE corresponds to approximately 3.76%–5.58% of the full score interval. This indicates that the model maintained relatively low score-level prediction errors in clinical laparoscopic cholecystectomy videos. Specifically, the MAE ranged from 0.752 to 1.103 for Calot Triangle Dissection and from 0.959 to 1.115 for Gallbladder Dissection, indicating limited fold-level variation in both phases. As shown in Fig. 6(b), the SRCC ranged from 0.618 to 0.840 for Calot Triangle Dissection and from 0.441 to 0.971 for Gallbladder Dissection. Compared with the robot-assisted training datasets, the SRCC values on HeiChole showed larger fold-level fluctuations. This may be related to the relatively small sample size and concentrated score distribution of the dataset. Since the observed skill scores in HeiChole are mainly distributed within the narrow range of 15–21, SRCC is more sensitive to local ranking changes and sample-level prediction errors.

Based on the HeiChole experiments, we further performed external validation on the private SHURUI robot-assisted cholecystectomy dataset. Specifically, the model trained on HeiChole was directly evaluated on SHURUI without any target-dataset fine-tuning. This setting was designed to assess the cross-dataset and cross-platform applicability of the proposed method when transferring from conventional laparoscopic cholecystectomy videos to real robot-assisted cholecystectomy videos. The model achieved MAE values of 1.453 and 1.582 for Calot Triangle Dissection and Gallbladder Dissection, respectively, with an average MAE of 1.518. Overall, the HeiChole results indicate that the proposed method maintained relatively low prediction errors and moderate ranking consistency across the two key phases of clinical laparoscopic cholecystectomy. The external SHURUI validation further suggests that, without target-domain fine-tuning, the model retained a certain level of score prediction capability on real robot-assisted clinical surgery data. Given the limited sample size of the SHURUI dataset, this result should be interpreted as preliminary evidence of cross-platform clinical applicability rather than a large-scale statistical conclusion.

2) *Results on the Private Hernia Repair Suturing Surgery Dataset:* To further validate the generalization capability of the proposed model, a private dataset comprising 34 hernia repair suturing surgery videos was constructed. Following the widely adopted GOALS evaluation standard for laparoscopic

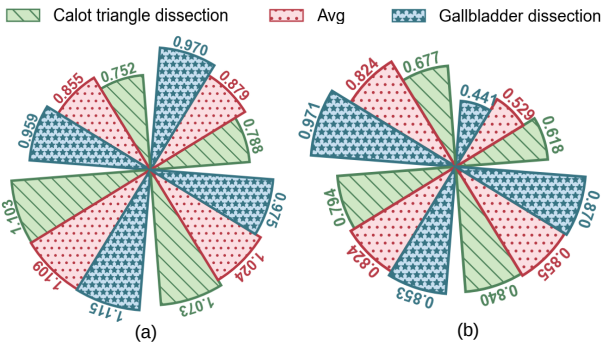


Fig. 6: Visualization of the results on the clinical cholecystectomy dataset. (a) Distribution of MAE under random 4-Fold cross-validation. (b) Distribution of SRCC under random 4-Fold cross-validation.

surgery, all videos were independently scored by experienced clinical experts to ensure annotation accuracy and consistency. Random 4-Fold cross-validation was employed to assess the robustness of the model. On this dataset, which has an average video duration of 188.1 seconds, the model achieved a mean MAE of 0.979 and a mean SRCC of 0.737 in the SSA task, demonstrating strong performance in both absolute score prediction and relative skill ranking. Specifically, as illustrated in Fig. 7, the MAE ranged from 0.806 to 1.067 (span: 0.261) across the four folds, representing only 4.03%–5.34% of the total scoring range, further confirming the high-precision regression capability of the model in clinical hernia repair suturing. Additionally, the SRCC ranged from 0.656 to 0.818 (span: 0.162), indicating stable performance in video skill ranking tasks. The observed superior performance can be primarily attributed to two factors. First, the contrastive learning framework effectively captures subtle skill differences among video samples. Second, by guiding the model to focus on the manipulation of surgical instruments in the foreground and on tissue state changes during surgery, accurate regression and reliable ranking are achieved even on a small dataset with a concentrated score distribution, such as the hernia repair videos.

Overall, the proposed framework exhibited a consistent performance pattern across different data sources and surgical scenarios. On the JIGSAWS robot-assisted training tasks, the model achieved high rank-order consistency and low prediction errors in most experimental settings, suggesting that query-reference contrastive modeling is effective for standardized robotic training scenarios. On the HeiChole and private hernia repair suturing datasets, SRCC showed larger variability, likely due to the limited sample size and concentrated score distributions; nevertheless, MAE remained low, indicating stable continuous score prediction in clinical laparoscopic videos. Furthermore, external validation on the SHURUI dataset showed that the model trained on conventional laparoscopic cholecystectomy videos could be directly transferred to real robot-assisted cholecystectomy videos without target-domain fine-tuning, while maintaining an acceptable prediction error. Taken together, these results support the applicability of the proposed framework for multi-source surgical video skill assessment and indicate that the complementary integration of instrument-region motion modeling and tissue-context representation contributes to video-level skill-score prediction.

C. Comparative Results

1) *Comparative Results on the JIGSAWS Dataset:* As shown in Table I, existing SSA methods can be broadly grouped by input modality into kinematic-based methods (K) [29], [55], video-based methods (V) [1], [11], and hybrid methods that integrate kinematic data and video (K+V) [9], [22]. Kinematic data provide direct measurements of instrument trajectories and motion patterns, but their availability often depends on specific robotic platforms and data interfaces. In contrast, endoscopic videos are more readily accessible across surgical scenarios. Therefore, achieving reliable skill

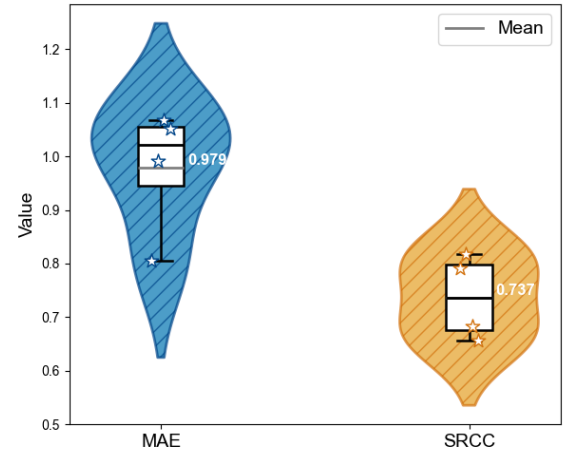


Fig. 7: Violin plots visualizing the MAE and SRCC distributions of the proposed model on the private hernia repair suturing dataset across 4-Fold cross-validation.

assessment using video alone is important for improving the cross-platform applicability of automated SSA systems. Overall, the proposed framework achieved competitive MAE and SRCC results across the three robot-assisted training tasks and three evaluation protocols on JIGSAWS. Compared with K and K+V methods, the proposed video-based framework obtained higher SRCC values in several cross-user settings. For example, under KT-LOUO, our method improved SRCC by 0.26 over the best K+V comparison method [9]. Under NP-LOUO, it achieved an SRCC of 0.94, outperforming the second-best K+V method [22]. These results indicate that query-reference contrastive modeling can effectively characterize relative skill differences between surgical videos without requiring kinematic inputs. Among video-based methods, ContraSformer [1] has demonstrated the effectiveness of reference-video-based contrastive regression for robotic surgical skill assessment. Building on this paradigm, the proposed framework further incorporates instrument-region motion modeling, segmentation-guided dynamic feature reweighting, and tissue-context representation. Compared with ContraSformer and other video-based approaches in Table I, our method achieved improved average MAE and SRCC, particularly under LOUO, where the model must generalize to unseen users. These results suggest that introducing fine-grained instrument-region and tissue-context cues into the reference-video contrastive regression framework facilitates video-level skill-score prediction.

2) *Comparative Results on the HeiChole Dataset:* To compare the proposed method with existing approaches in clinical cholecystectomy scenarios, we conducted a systematic evaluation on the HeiChole dataset, as summarized in Table III. Compared with prior methods, the proposed framework achieved lower score prediction error and higher rank-order consistency, with an MAE of 0.97 and an SRCC of 0.76. Relative to the best-performing comparison method, MT-ViT [11], the MAE decreased from 1.06 to 0.97, while the SRCC increased from 0.58 to 0.76. These results indicate that the proposed framework improves both continuous score regression

TABLE III: Quantitative comparison results on the clinical cholecystectomy dataset, **values in parentheses** indicate the percentage improvement over the best baseline.

Method	Metrics	
	MAE	SRCC
MUSDL [33]	1.58	0.23
Multiph-VTE [22]	1.34	0.35
VISA [21]	1.27	0.46
Contra-Sformer [1]	1.33	0.41
MT-VIT [11]	1.06	0.58
Ours	0.97 (+8.49%)	0.76 (+31.03%)

and relative skill ranking on clinical laparoscopic cholecystectomy videos. These performance gains are primarily attributed to the synergistic enhancement of the model’s perceptual and discriminative capabilities. On the one hand, the model effectively focuses on key skill-related dimensions, such as instrument manipulation and tissue state variations during surgery, and extracts fine-grained features from these cues. On the other hand, the introduced contrastive learning framework constructs fine-grained comparisons of instrument operations and tissue changes between reference and query videos, further enhancing the model’s sensitivity and discriminative power to subtle inter-video differences, thereby enabling more robust and reliable skill differentiation and ranking.

D. Ablation Results

To validate the effectiveness of each proposed component, a series of experiments were performed under the 4-fold cross-validation setting on the JIGSAWS dataset, as summarized in Table IV. In Table IV, Backbone denotes the I3D-based dynamic branch for spatiotemporal feature extraction, Instrument represents the segmentation-guided feature reweighting module, Tissue refers to the ResNet-34-based tissue-context branch, and Contra indicates the query-reference contrastive regression component. Seven ablation experiments were designed to analyze the influence of different component combinations on model performance. It should be noted that when Contra is not incorporated, the model directly regresses the annotated skill scores from the extracted features rather than estimating the score difference relative to a reference video. The experimental results indicate that when the model relies solely on single-video features for SSA, the overall SRCC remains within the range of 0.396–0.517. After the query-reference contrastive regression component is introduced, the SRCC range increases to 0.600–0.880, indicating that explicitly modeling relative differences between query and reference videos improves ranking consistency. Regardless of whether the contrastive regression component is employed, removal of the Instrument module (e.g., comparing E and A with F and B) results in a substantial decline in both MAE and SRCC. A similar performance degradation is observed when the Tissue module is removed. These results indicate that instrument-region motion features and tissue-context features provide complementary information for SSA and jointly improve both score prediction accuracy and ranking consistency.

V. DISCUSSIONS

A. Evaluation On The Selection Of Reference Video

Following the reference-video selection analysis in Contra-Sformer [1], we evaluated how the choice of reference video affects the performance of the proposed framework. Specifically, 4-fold cross-validation experiments were conducted on the three JIGSAWS tasks. For each task, videos with low, intermediate, and high annotated skill scores were selected as reference videos, and the results are summarized in Table V. The results show that reference video selection has a clear effect on model performance. Across all three tasks, using the highest-scoring video as the reference consistently outperformed using low- or intermediate-scoring reference videos. On average, high-scoring references reduced MAE by 1.090 and 1.018 compared with low- and intermediate-scoring references, respectively, and improved SRCC by 40.80% and 26.80%. This finding is consistent with the role of high-scoring videos as more stable reference samples. High-scoring surgical videos generally exhibit more standardized procedural execution, more continuous instrument motion, and fewer non-informative actions, thereby providing a more consistent reference for query videos. In contrast, low-scoring videos often contain more diverse operational deviations, and their error patterns may vary across samples, which can reduce their representativeness as reference videos. Based on these results, the highest-scoring video was used as the reference video in the other experiments

B. Evaluation On The Number Of Attention Heads

To evaluate the influence of the number of heads in the cross-attention decoders, we conducted an ablation study under the randomized 4-fold cross-validation setting on the JIGSAWS dataset, as reported in Table VI. The number of heads controls how the fixed embedding dimension is divided into multiple attention subspaces, and therefore affects the balance between multi-subspace representation and the feature capacity assigned to each head. The results show that the model achieved the best average performance when the number of heads was set to 8, with a mean MAE of 2.171 and an SRCC of 0.880. When only one head was used, the model reduced to a single attention space, and the SRCC decreased to 0.524. This suggests that a single attention subspace is insufficient to capture the heterogeneous visual dependencies involved in surgical skill assessment. Increasing the number of heads from 2 to 4 and then to 8 improved the average performance, indicating that multiple attention subspaces can provide complementary representations for query-reference comparison. However, further increasing the number of heads to 16 or 32 degraded performance, with the average SRCC decreasing to 0.562 and 0.600, respectively. This result indicates that more heads do not necessarily lead to better performance. Under a fixed embedding dimension, increasing the number of heads reduces the feature dimension assigned to each head, which may weaken the representation capacity of each attention subspace and make the learned attention patterns less stable. In addition, excessive partitioning may disperse attention across redundant or weakly informative subspaces,

TABLE IV: Quantitative ablation results of the proposed model on the simulated dataset.

Experiments	Components				Tasks							
	Backbone	Instrument	Tissue	Contra	KT		NP		SU		Avg.	
					MAE	SRCC	MAE	SRCC	MAE	SRCC	MAE	SRCC
A	✓				3.946	0.357	3.627	0.422	4.338	0.408	3.970	0.396
B	✓	✓			4.001	0.497	3.242	0.517	3.909	0.411	3.717	0.475
C	✓		✓		3.622	0.505	3.439	0.493	4.219	0.404	3.760	0.467
D	✓	✓	✓		3.520	0.472	3.187	0.545	3.779	0.533	3.495	0.517
E	✓			✓	3.068	0.650	2.942	0.703	3.845	0.446	3.285	0.600
F	✓	✓		✓	2.947	0.624	3.120	0.622	3.096	0.694	3.139	0.647
G	✓		✓	✓	2.561	0.814	3.338	0.681	3.349	0.526	3.083	0.674
Ours	✓	✓	✓	✓	1.918	0.889	1.811	0.893	2.785	0.857	2.171	0.880

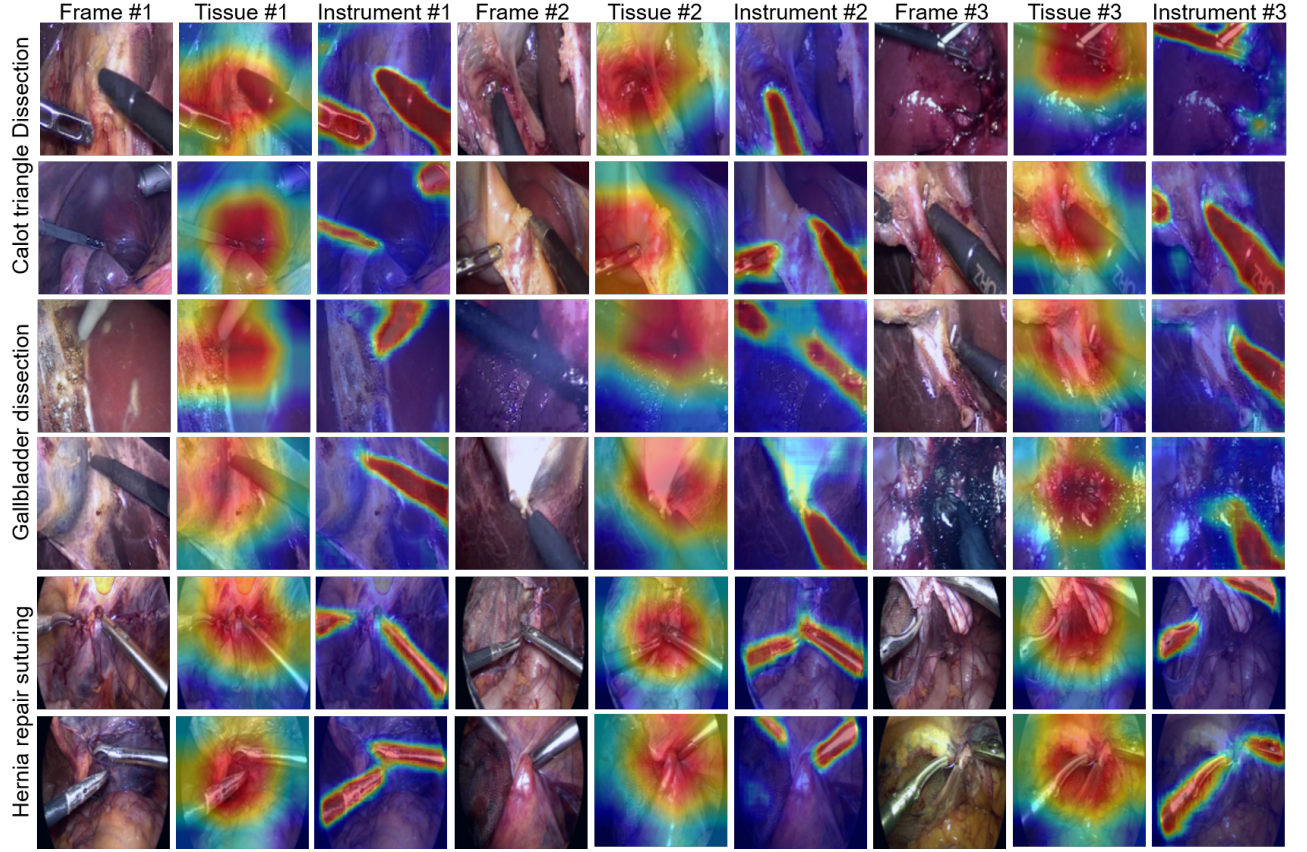


Fig. 8: The CAM visualizations of instrument features extracted by the mask-guided attention module and tissue features extracted by the intra-video contextual modeling module are shown for the clinical dataset.

TABLE V: Evaluation on the selection of reference video on the simulated dataset.

Ref	KT		NP		SU		Avg.	
	MAE	SRCC	MAE	SRCC	MAE	SRCC	MAE	SRCC
Lowest	3.152	0.694	3.082	0.661	3.550	0.519	3.261	0.625
Intern	2.937	0.680	2.944	0.794	3.687	0.607	3.189	0.694
Highest	1.918	0.889	1.811	0.893	2.785	0.857	2.171	0.880

making optimization more difficult. Therefore, the observed performance drop from 8 to 16 heads reflects the sensitivity of the proposed framework to the allocation of attention

TABLE VI: Evaluation on the Number of Attention Heads on the simulated dataset.

Heads	KT		NP		SU		Avg.	
	MAE	SRCC	MAE	SRCC	MAE	SRCC	MAE	SRCC
1	3.777	0.397	2.924	0.756	3.641	0.418	3.447	0.524
2	3.705	0.449	3.433	0.726	3.503	0.565	3.547	0.580
4	3.601	0.612	3.167	0.802	3.415	0.615	3.394	0.676
8	1.918	0.889	1.811	0.893	2.785	0.857	2.171	0.880
16	3.333	0.449	3.004	0.712	3.337	0.526	3.225	0.562
32	3.407	0.551	3.268	0.677	3.278	0.571	3.318	0.600

capacity, rather than a monotonic relationship between the number of heads and performance. In our experiments, 8 heads provided the best trade-off between representation diversity and per-head feature capacity, and this setting was used in the subsequent experiments.

C. CAM Visualization of Tissue and Instrument Features

To qualitatively examine the visual regions emphasized by the model, we employed CAM [54] as a post hoc visualization tool. As shown in Fig. 8, representative frames from cholecystectomy and hernia repair suturing videos show that the activated regions are mainly located around surgical instruments, adjacent tissue structures, and instrument-tissue interaction areas. In cases with visual disturbances such as smoke or bleeding, the CAM responses still tend to cover regions associated with the ongoing manipulation. These visualizations provide qualitative support that the proposed framework learns local skill-related visual context, which is consistent with the design of instrument-region motion modeling and tissue-context representation. More complete video-level CAM visualizations will be made available on the project homepage.

D. Scoring Reliability and Annotation Consistency

Surgical skill assessment relies heavily on expert annotation, and label quality directly affects model training, performance evaluation, and result interpretation. Therefore, beyond model performance, the reliability of the scoring instruments and the agreement among raters are important considerations GOALS. OSATS is a widely used structured tool for evaluating surgical technical performance, including tissue handling, instrument handling, flow of operation, and overall performance. Previous studies have shown that OSATS-based global rating scales can distinguish surgeons at different training levels and provide good internal consistency and inter-rater reliability [25], [34]. Similarly, GOALS is a global rating tool developed for laparoscopic skill assessment, covering key dimensions such as depth perception, bimanual dexterity, efficiency, and tissue handling. Prior studies have also demonstrated that GOALS can differentiate operators with different experience levels and has good validity and reliability in laparoscopic cholecystectomy training settings [38], [40]. To improve the reliability of the labels for the newly incorporated datasets, both the ROSMA dataset and the private hernia repair suturing dataset were independently scored by surgeons with extensive clinical experience. When scoring discrepancies occurred, the videos were reviewed and discussed to obtain the final annotations.

For the ROSMA dataset, two surgeons independently scored the three robot-assisted training tasks: Post and Sleeve, Pea on a Peg, and Wire Chaser. Annotation consistency was evaluated at the total-score level using ICC(2,1), which corresponds to a two-way random-effects, absolute-agreement, single-rater intraclass correlation coefficient [42]. Across the three tasks, the ICC(2,1) values between the two surgeons and between each surgeon and the final labels were all above 0.90, indicating high absolute agreement among different scoring sources. These results suggest that the re-annotated ROSMA labels

are reliable and provide stable supervision signals for model training and evaluation in robot-assisted training tasks. For the private hernia repair suturing dataset, two surgeons with extensive clinical experience independently scored 34 laparoscopic suturing videos. The total-score-level agreement analysis showed an ICC(2,1) of 0.91 between the two surgeons, indicating high consistency in their independent evaluations. Overall, OSATS and GOALS provide structured and widely adopted scoring frameworks for surgical skill assessment. For the newly incorporated ROSMA and private hernia repair suturing datasets, we further assessed annotation consistency at the total-score level using ICC(2,1). The results indicate that these annotations have high inter-rater agreement, providing relatively reliable supervision signals for the proposed video-based surgical skill assessment framework.

E. Multi-Source Validation for Surgical Skill Assessment

VI. CONCLUSIONS

Surgical videos are primarily composed of foreground instrument manipulations, midground tissue structures, and background anatomical context. Accordingly, we designate instrument operation, tissue handling, and surgical field clarity as the key dimensions for assessing surgical skill. Furthermore, since single-video-based skill score regression methods often fail to capture subtle inter-video variations, we propose a fine-grained contrastive learning framework to enable precise and reliable SSA. Specifically, spatiotemporal features are first extracted from surgical videos using the I3D network as the backbone. A mask-guided attention module is subsequently applied to highlight critical instrument regions in the video frames. These features are fused with the spatiotemporal representations to produce dynamic instrument-operation features. Subsequently, an intra-video contextual modeling module, built upon ResNet, extracts static visual features from the video frames, effectively capturing tissue state changes and variations in surgical field clarity throughout the procedure. Cross-attention mechanisms are employed to derive differential features between the query and reference videos, focusing on dynamic instrument manipulation and tissue state changes. Finally, the differential features are input into a multilayer perceptron, where the score differences between videos are regressed using a mean squared error loss function, thereby enabling precise SSA.

To systematically evaluate the effectiveness and generalizability of the proposed model, extensive experiments were conducted on both simulated and clinical datasets (laparoscopic cholecystectomy and hernia repair suturing). Notably, the model achieved state-of-the-art performance on both MAE and SRCC metrics across both the simulated dataset and the cholecystectomy dataset. First, we conducted experiments on the simulated dataset, which contains three fundamental surgical tasks (KT, NP, and SU), under three experimental settings: LOSO, LOUO, and random 4-Fold cross-validation. The model achieved MAE values of 1.92, 1.78, and 2.17, and SRCC values of 0.91, 0.84, and 0.88 under the three respective settings. These results indicate that the model can not only accurately regress surgical skill scores but also effectively

capture subtle inter-video differences, enabling reliable skill ranking. On the cholecystectomy dataset, the model achieved an MAE of 0.97 and an SRCC of 0.76, demonstrating strong generalization ability and potential clinical applicability. To further assess performance on different types of clinical surgeries, a private dataset containing 34 hernia repair suturing videos was collected, with independent scoring performed by two surgeons using the GOALS assessment criteria specific to laparoscopic surgery. On this private hernia repair dataset, the model achieved an MAE of 0.98 and an SRCC of 0.74, further confirming its reliability in clinical scenarios. Moreover, a total of 169,139 high-quality surgical instrument masks, constituting the largest dataset currently available for SSA tasks and encompassing both simulated and clinical videos, were generated using SAM2, providing a reliable and comprehensive data support for future research.

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